

# Quality Control Standards — Helix Pharmaceuticals Building 4

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|----------------|----------------------------------|----------------|----------------------|
| Document ID    | HLX-QC-001                       | Revision       | 2                    |
| Plant          | Helix Pharmaceuticals Building 4 | Effective Date | 20 Mar 2026          |
| Document Owner | L. Hartmann, Head of Quality     | Classification | Controlled — Quality |

## 1. Purpose & Scope

This standard defines the QC strategy for sterile drug products manufactured at Helix Pharmaceuticals Building 4 (small-molecule and ADC). It governs raw material release, in-process control, finished product release, and stability testing under FDA 21 CFR 211 and ICH guidelines.

## 2. Product Specifications

| Product                     | Key Attributes                                        | Specification Reference                                   |
|-----------------------------|-------------------------------------------------------|-----------------------------------------------------------|
| API                         | Identity, assay, related substances, water            | USP/EP method; assay 98.0-102.0%; impurities per ICH Q3A. |
| Drug product (sterile vial) | Identity, assay, impurities, sterility, endotoxin, pH | USP/EP method; assay 95-105%.                             |
| ADC bulk                    | Conjugation DAR, free drug, aggregation               | DAR by HIC/UV; aggregation by SEC $\leq 2.0\%$ .          |

## 3. Procedures

### 3.1 Raw Material Release

Each lot of API and excipient tested per the approved specification. CoA received from supplier verified against on-site testing. Released lots tagged with QA-released status in the eQMS.

### 3.2 In-Process Control

Critical Process Parameters (CPPs) monitored in real time. Out-of-trend events trigger a deviation; the batch may proceed only after Quality on Floor decision.

### 3.3 Stability

ICH Q1A long-term and accelerated stability per program PRG-HLX-STAB-01. Sample pulls scheduled at 0, 3, 6, 9, 12, 18, 24, and 36 months.

### 3.4 Data Integrity

All QC data under 21 CFR Part 11; audit trails reviewed by QA on a defined frequency. ALCOA+ principles applied to manual and electronic records.

## 4. Out-of-Specification Handling

Any OOS result triggers an immediate Hold on the affected lot, a documented investigation, and where applicable, a CAPA. Disposition decisions (Release, Rework, Reject) require QA approval. The investigation is closed in the eQMS with full traceability per the site's records retention policy.

## 5. Roles & Responsibilities

Quality Assurance (L. Hartmann, Head of Quality) owns lot release. Operations executes per the recipe and reports deviations. Maintenance supports calibration of QC-critical instruments per the linked maintenance manual (HLX-MNT-001). The site safety procedure (HLX-SAF-001) applies to all QC laboratory activity.

## 6. References

- ICH Q1A(R2), Q3A, Q3B, Q6A
- USP <71>, <85>, <788>
- FDA 21 CFR Part 211 Subpart I
- Internal: PRG-HLX-STAB-01, eQMS QA-HLX-101

## 7. Revision History

| Rev | Date        | Author      | Summary                                  |
|-----|-------------|-------------|------------------------------------------|
| 1   | 18 Sep 2024 | L. Hartmann | Initial issue.                           |
| 2   | 20 Mar 2026 | L. Hartmann | Updated specifications and SPC criteria. |

### Approvals

| Role                    | Name                         | Signature | Date        |
|-------------------------|------------------------------|-----------|-------------|
| Quality Assurance       | L. Hartmann, Head of Quality | _____     | 20 Mar 2026 |
| Operations / Site Lead  | A. Nakamura, Site Head       | _____     | 20 Mar 2026 |
| Regulatory / Compliance | Site Compliance Officer      | _____     | 20 Mar 2026 |

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